

LOGITHKUMAR K R

Software Engineer | AI Integrations, Backend & Full-Stack Systems
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Executive Summary

Software Development Engineer with hands-on experience building and deploying full-stack applications integrated with AI and machine learning capabilities, including two systems live in production at Bannari Amman Institute of Technology. Proficient in designing scalable backend systems, RESTful APIs, and real-time data pipelines, with practical expertise in LLM integration and constraint optimization. Passionate about applying system design principles and object-oriented architecture to ship production-grade software that solves real-world problems.

Projects

College Timetable Generator - CP-SAT Constraint Scheduling System

Constraint Optimization & Full-Stack Systems

Feb 2026 - Present

- Engineered a **CP-SAT constraint solver** (Google OR-Tools) that auto-generates **conflict-free college timetables**, enforcing faculty clash prevention, mentor-hour blocking, and lab batch rotation across a 6-day × 8-period grid.
- Built a **ConstraintInterpreter** that compiles user-defined JSON rules into solver constraints pre-solve and injects custom sessions post-solve — enabling admins to configure scheduling policies without touching code.
- Designed a **multi-section dynamic enrollment engine** that calculates section counts per course from **live enrollment data** and auto-assigns venues based on capacity — eliminating manual scheduling for large departments.
- Tech Stack:** Python, FastAPI, Google OR-Tools (CP-SAT), SQLAlchemy, SQLite, React (Vite), dnd-kit, Tailwind CSS, JWT.
- Live:** timetable.bitsathy.ac.in **Code:** github.com/logithKR/Time-table-generator

BITRA - Institutional RAG Chatbot & Knowledge Assistant

Generative AI & NLP Systems

Oct 2025 - Dec 2025

- Built a **RAG chatbot** using **LangChain + FAISS + Gemini 2.5 Flash** with a two-stage LLM pipeline — classifier routes queries to 5 departments; RAG chain delivers context-grounded answers with hallucination prevention.
- Implemented **Gmail API (OAuth 2.0) auto-escalation** to department heads and a **live FAISS index rebuild** system enabling real-time knowledge base updates via a staff document portal.
- Tech Stack:** Python, Flask, LangChain, FAISS, Gemini 2.5 Flash, Gmail API, Firebase Auth, React (Vite).
- Live:** bitra.bitsathy.ac.in

GenJewels - Generative AI Jewelry Design Platform

Generative AI & Diffusion Models

Jul 2025 - Sep 2025

- Built a **generative AI platform for jewelry** design featuring two core pipelines: **Text-to-Image** for creating custom designs from descriptive prompts, and **Image-to-Image** for applying AI style transfers to user-uploaded photos.
- Integrated a **Stable Diffusion pipeline** utilizing **dynamic prompt engineering** and **denoising strength controls** to accurately process source images and user parameters into high-fidelity, photorealistic renders.
- Tech Stack:** React (Vite), Tailwind CSS, Stable Diffusion API.
- Code:** github.com/Ramesh-Sakthivel-18/Gen-Jewels

Educational Background

Bannari Amman Institute of Technology

B.E - Computer science and design

Sep 2023 - May 2027

CGPA - **8.14** / 10

URC Palaniammal Matriculation Higher Secondary School

High School - Math Computer science

Jun 2021 - May 2023

93.67%

Technical Skills

- Languages:** Python, JavaScript, SQL
- Frontend:** React (Vite), Tailwind CSS, HTML, CSS
- Backend:** FastAPI, Flask, Node.js, Express.js, REST APIs, WebSockets
- Databases:** SQLite, Firebase Firestore, Redis, SQLAlchemy
- AI/ML:** LangChain, FAISS, Google Gemini API, Google OR-Tools (CP-SAT), Stable Diffusion API
- Tools & Auth:** Git, GitHub, JWT, Firebase Auth, OAuth 2.0, Socket.io

Awards & Honors

Reliance Foundation Undergraduate Scholar 2023

Awarded the Reliance Foundation Undergraduate Scholarship for academic excellence, selected as one of 5,000 scholars across India.

Selected among the Top 7 teams in INTELLIMOBILITY – Driving Safe Mobility Ideathon conducted by Automotive Research Association of India (ARAI), Pune, for presenting an AI-based pedestrian detection and smart safety system.

Certifications

- AI Fundamentals with IBM SkillsBuild — Cisco Networking Academy, IBM
- Programming, Data Structures and Algorithms using Python — NPTEL

Online Profiles

 [GitHub - logithKR](https://github.com/logithKR)

 [LinkedIn - LOGITHKUMAR K R](https://www.linkedin.com/in/logithkumar188)

Spoken Languages

- English
- Tamil